How Order-Sensitive Are LLMs?
OrderProbe for Deterministic Structural Reconstruction

Yingjie He[*,1], **Zhaolu Kang**[*,1], **Kehan Jiang**[*,1], **Qianyuan Zhang**[2], **Jiachen Qian**[3], **Chunlei Meng**[4], **Yujie Feng**[5], **Yuan Wang**[6], **Jiabao Dou**[1], **Aming Wu**[1], **Leqi Zheng**[7], **Pengxiang Zhao**[6], **Jiaxin Liu**[8], **Zeyu Zhang**[1], **Lei Wang**[1], **Guansu Wang**[1], **Qishi Zhan**[1], **Xiaomin He**[1], **Meisheng Zhang**[1], **Jianyuan Ni**[†,9],
[1]Peking University [2]The Chinese University of Hong Kong, Shenzhen
[3]City University of Hong Kong [4]Fudan University [5]The Hong Kong Polytechnic University
[7]Tsinghua University [6]Zhejiang University [8]University of Illinois Urbana-Champaign
[9]Marquette University [10]Juniata College

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Abstract

Large language models (LLMs) excel at semantic understanding, yet their ability to reconstruct internal structure from scrambled inputs remains underexplored. Sentence-level restoration is ill-posed for automated evaluation because multiple valid word orders often exist. We introduce **OrderProbe**, a deterministic benchmark for structural reconstruction using fixed four-character expressions in Chinese, Japanese, and Korean, which have a unique canonical order and thus support exact-match scoring. We further propose a diagnostic framework that evaluates models beyond recovery accuracy, including semantic fidelity, logical validity, consistency, robustness sensitivity, and information density. Experiments on twelve widely used LLMs show that structural reconstruction remains difficult even for frontier systems: zero-shot recovery frequently falls below 35%. We also observe a consistent dissociation between semantic recall and structural planning, suggesting that structural robustness is not an automatic byproduct of semantic competence. To facilitate reproducibility, we release OrderProbe at https://github.com/Zhaolu-K/OrderProbe.

1 Introduction

Large language models (LLMs) have demonstrated strong performance on semantic understanding and generation tasks. However, their robustness to internal structural disruption remains an open question (OpenAI et al., 2024; DeepSeek-AI et al., 2025). In human language processing, listeners can often recover meaning from structurally disordered inputs with little effort. Replicating this evaluation for machine models, however, poses a fundamental methodological challenge. The flexibility of natural language syntax means that a scrambled sentence often admits multiple valid permutations, preventing a unique ground truth and making deterministic evaluation difficult. This ambiguity limits sentence-level restoration benchmarks and often conflates structural ability with general fluency.

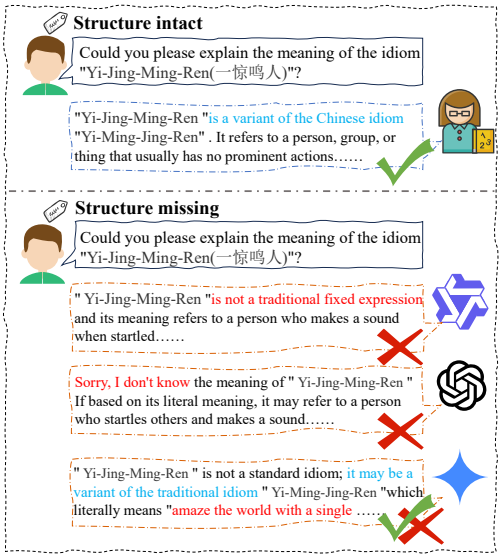

Figure 1: **OrderProbe example illustrating semantic–structure dissociation.** Given a scrambled four-character expression, humans can often recover the unique canonical order using semantic anchors and structural priors. Many LLMs, however, produce correct meanings while failing exact reconstruction, revealing a gap between semantic recall and structural planning.

Existing robustness evaluations largely target surface noise or external instruction perturbations rather than internal ordering constraints(Wang et al., 2021; Ribeiro et al., 2020; Li et al., 2024). Although these benchmarks have advanced robustness evaluation, they do not directly probe whether a model can reconstruct a globally constrained sequence from scrambled components. This omission is particularly important given evidence that Transformer models can exhibit limited sensitivity

[*]Equal contribution.
[†]Corresponding author.

Benchmark	Year	Language	Target Unit	Perturbation Level	Core Task
General Robustness Benchmarks					
RoTBench (Ye et al., 2024)	2024	En	Tool-use I/O	Real-world Noise	Tool-use Robustness (Tool Learning)
Instruct-Robust (Li et al., 2024)	2024	En	Instructions	External Injection	Instruction-Following Robustness
ReDial (Lin et al., 2025)	2025	En (Dialect)	Logic Problems	Dialectal Variation	Reasoning (Math/Logic)
Idiom & Figurative Language Benchmarks					
MMFLD (Lai et al., 2023)	2023	Multi	Fig. Language	*None* (Static)	Detection / Class.
WenMind (Cao et al., 2024)	2024	Zh	Classical Text	*None* (Static)	QA & Generation
Chengyu-Bench (Fu et al., 2025)	2025	Zh	Idioms	*None* (Static)	Understanding & Use
MChIRC (Wang et al., 2025)	2025	Zh	Idiom-in-Context	*None* (Static)	Multimodal RC / Cloze
Idiom-Trans (Donthi et al., 2025)	2025	Multi	Idioms	*None* (Static)	Translation
OrderProbe (Ours)	**2025**	**CJK**	**Idioms**	**Internal Structure**	**Reconstruction**

Table 1: **Positioning of OrderProbe.** Existing robustness benchmarks mainly test surface-level noise or instruction-level perturbations, while idiom benchmarks treat expressions as static retrieval or transformation targets. OrderProbe uniquely evaluates deterministic internal-order reconstruction with exact-match scoring.

to word order (Sinha et al., 2021), and may fail to generalize structural relations under reversed or reordered contexts (Berglund et al., 2024).

To isolate structural reconstruction under deterministic scoring, we use fixed four-character expressions in Chinese, Japanese, and Korean as the evaluation unit. Sentence-level restoration is ill-posed because scrambled sentences often admit multiple valid reorderings, preventing a unique ground truth. In contrast, four-character expressions typically have a rigid internal sequence and a single canonical form, enabling exact-match evaluation while preserving high semantic density in a minimal structure. This setting also differs from existing idiom and figurative language benchmarks, which mainly test retrieval or interpretation under canonical inputs (Zheng et al., 2019; Lai et al., 2023; Fu et al., 2025) or translation and usage (Donthi et al., 2025), rather than reconstruction from scrambled constituents. Classical understanding datasets similarly emphasize semantics without probing structure recovery under internal perturbation (Cao et al., 2024).

Figure 6 illustrates this distinction. When presented with scrambled characters from the idiom *Yi-Ming-Jing-Ren* ("to become famous overnight"), humans can quickly reconstruct the canonical order based on semantic anchors. By contrast, although LLMs can provide accurate definitions when queried directly, they often fail to recover the correct structure when constrained to use the scrambled components. Instead, models tend to process the characters as independent tokens. This behavior reveals a dissociation where models memorize the semantic entry of an expression but fail to reassemble its canonical structure from its parts.

To systematically evaluate this dissociation, we introduce OrderProbe, a benchmark designed to measure structural reconstruction capabilities in Chinese, Japanese, and Korean. The dataset is constructed through a rigorous pipeline involving multi-source acquisition, expert filtering, and hybrid semantic annotation to ensure linguistic precision. It comprises 3,543 curated samples, covering six syntactic categories and multiple script typologies. Beyond dataset construction, we propose a diagnostic evaluation framework that decomposes model behavior into six dimensions, including reconstruction accuracy, semantic fidelity, logical validity, structural consistency, robustness sensitivity, and information density. Unlike approaches that rely solely on coarse accuracy rankings, this framework enables fine-grained analysis of failure modes that separate semantic recall from structural planning.

Our contributions can be summarized as follows.

- We identify the ambiguity of sentence-level restoration and establish fixed four-character expressions as a deterministic proxy for evaluating structural reconstruction.
- We introduce OrderProbe, a multilingual benchmark comprising 3,543 curated samples across six syntactic categories.
- We propose a diagnostic evaluation framework that moves beyond simple scoring to characterize robustness sensitivity, semantic fidelity, and hallucination behavior.
- We evaluate twelve widely used LLMs on OrderProbe and find that exact reconstruction remains challenging, with zero-shot recovery frequently below 35% even for state-of-the-art systems.

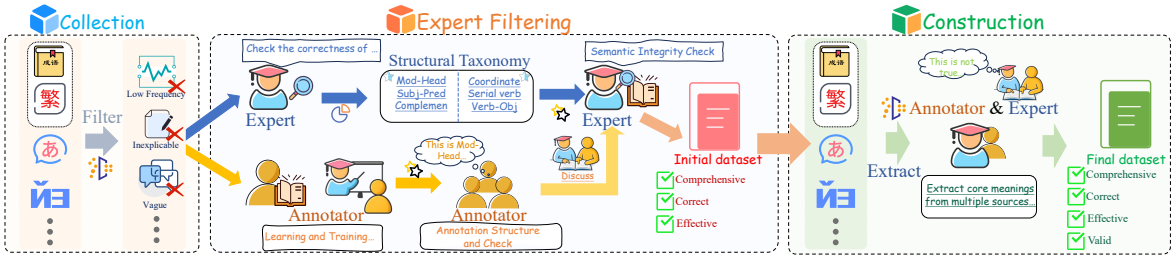

Figure 2: **OrderProbe construction pipeline.** We collect raw four-character expressions from multiple sources, apply expert filtering to obtain canonical roots, build semantic references via dictionary grounding and verified augmentation, and generate the full non-identity permutation space as structurally perturbed evaluation inputs.

2 OrderProbe

OrderProbe is constructed through a controlled pipeline that transforms raw lexical resources into a deterministic benchmark for structural reconstruction. The pipeline consists of four stages: multi-source collection, expert filtering, semantic reference construction, and permutation-based perturbation generation. This section describes each stage and summarizes the resulting dataset.

2.1 Stage I: Multi-source Collection

We first assemble a candidate pool $\mathcal{C}_{raw}$ by aggregating four-character expressions from multiple publicly available resources, including lexical dictionaries and digital repositories. Each candidate item is associated with a script typology $T \in \{\text{ZH-CN}, \text{ZH-TW}, \text{JA}, \text{KO}\}$. The goal of this stage is to maximize coverage and diversity before quality control. This process yields approximately 4,000 raw candidates.

2.2 Stage II: Expert Filtering

To obtain a linguistically reliable core set, we apply an expert filtering function $\mathcal{F}$ that maps $\mathcal{C}_{raw}$ to a refined set $\mathcal{X}$:

$$\mathcal{X} = \mathcal{F}(\mathcal{C}_{raw}). \tag{1}$$

Filtering is performed by senior linguists and trained annotators with proficiency in the target languages. Items are removed if they are non-standard, ambiguous, modern internet slang, or exhibit multiple competing canonical forms. After cross-lingual alignment and consensus review, this process yields exactly 3,543 canonical expressions in $\mathcal{X}$, which form the benchmark roots.

2.3 Stage III: Semantic Reference Construction

For each canonical expression $x \in \mathcal{X}$, we construct a semantic reference set $\mathcal{S}_x$ to support evaluation beyond exact recovery. We retrieve dictionary definitions d_{dict} as the primary semantic anchor, following common practice in idiom evaluation (Zheng et al., 2019; Fu et al., 2025). To reduce brittleness to surface phrasing, we augment each definition with paraphrastic variants generated by LLMs and verified by annotators, similar to recent benchmark construction protocols (Li et al., 2024). The final reference set is defined as:

$$\mathcal{S}_x = \{d_{dict}\} \cup \{d_{aug}\}. \tag{2}$$

This hybrid construction enables robust semantic and logical evaluation without translating across languages.

2.4 Stage IV: Permutation-based Perturbation Generation

The benchmark evaluates whether models can reconstruct the canonical order of an expression from scrambled constituents. Each canonical expression x is represented as a token sequence (t_1, t_2, t_3, t_4) with a fixed order. We apply a perturbation operator $\mathcal{P}$ that generates all non-identity permutations:

$$\mathcal{P}(x) = \{\pi(x) \mid \pi \in S_4, \pi(x) \neq x\}, \tag{3}$$

where S_4 denotes the permutation group over four elements. This yields 23 perturbed inputs per canonical expression, forming a complete combinatorial space for internal reordering. Unlike surface perturbations such as typos or synonym substitutions (Wang et al., 2021; Ribeiro et al., 2020), this operator preserves all lexical content while disrupting only the internal structure.

2.5 Dataset Statistics

The final dataset $\mathcal{D}$ consists of all perturbed variants generated from the canonical set $\mathcal{X}$. It covers four script settings (Simplified Chinese, Traditional Chinese, Japanese, and Korean) and six syntactic